4

Principal Component Analysis and Unsupervised Pattern Recognition

4.1 Introduction

Multivariate analysis is the core of chemometrics, and most of the first papers in the literature in the 1970s used simple multivariate methods to explore complex chemical data sets. The key concept is that many studies in chemistry and associate laboratory measurements (e.g. in biology or medicine) involve recording many variables per sample. For example, a spectrum may be recorded at hundreds of wavelengths, or a chromatogram at many elution times. Hence, for each sample, we have a lot of information. Traditionally, one or two variables (e.g. the intensity at characteristic wavelengths) were used to describe a sample. However, this poses the problem of choosing the best variables, and often, there is a lot of information mixed up in the signal. A mixture may consist of half a dozen or even more compounds: most materials of interest, whether the extract from a plant, or a fruit juice, or a piece of wood or a bodily fluid, consist, in fact, of many compounds. They may not all absorb at a single spectroscopic wavelength; hence, the composition cannot necessarily be obtained just by monitoring single wavelengths. Often, we do not want to know the full chemical composition of materials: this may be very complicated and time consuming and the aim of an investigation might be primarily to identify, for example, the origin of the sample: samples of similar provenance may group together; hence, we may just want to visualise how our sample fits in to an existing set.

There are numerous ways we might want to deal with this multivariate data. The methods described in this chapter are unsupervised; that is, they do not assume any pre-defined pattern or grouping and usually result in visualisation in the form of graphs. Common to all these approaches, data are represented in matrix form, usually with the rows corresponding to single samples (or observations, e.g., a spectrum or a chromatogram) and columns to variables (e.g. wavelengths or elemental composition) and the cells usually equalling intensity.

We will discuss two main groups of techniques in this chapter.

4.1.1 Exploratory Data Analysis

Exploratory data analysis (EDA) mainly consists of principal components analysis (PCA) and factor analysis (FA). Classical statistical examples come from biology and psychology. Psychometricians have, for many years, had the need to translate numbers, such as answers to questions in tests, into relationships between individuals. How verbal ability, numeracy and the ability to think in three dimensions can be predicted from a test? Can different people be grouped by these abilities? And does this grouping reflect the backgrounds of the people taking the test? Are there differences according to educational background, age, sex or even linguistic group?

In chemistry, we too need to ask similar questions, but the raw data are often chromatographic or spectroscopic. An example is animal pheromones: animals recognise each other more by smell than by sight, and different animals often lay scent trails, sometimes in their urine. The chromatogram of a urine sample may contain several hundred compounds, and it is often not obvious to the untrained observer to decide which samples are most significant. Sometimes, the most potent compounds are only present in small quantities. Yet, animals can often detect through scent marking that whether there is one in-heat member of the opposite sex looking for a mate, or whether there is a dangerous intruder entering their territory. EDA of chromatograms of urine samples can highlight differences in chromatograms of different social groups or different sexes and gives a simple visual idea to the main relationships between these samples. Sections 4.2–4.6 cover these approaches.

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4.1.2 Cluster Analysis

A more formal method of treating samples is cluster analysis, the most widespread method for unsupervised pattern recognition in chemometrics. Many methods have their origins in numerical taxonomy. Biologists measure features in different organisms, for example, various body length parameters. Using a couple of dozen features, it is possible to see which species are most similar and draw a picture of these similarities, called a *dendrogram*, in which more closely related species are closer to each other. The main branches of the dendrogram can represent bigger divisions, such as subspecies, species, genera and families.

These principles can be directly applied to chemistry. It is possible to determine similarities in amino acid sequences of myoglobin in a variety of species. The more similar the species, the closer the relationship: chemical similarity mirrors biological similarity. Sometimes, the amount of information is huge; for example, in large genomic or crystallographic databases, cluster analysis is the only practicable way of searching for similarities.

Unsupervised pattern recognition differs from EDA in that the aim of the methods is to detect similarities between species, whereas using EDA, there is no particular prejudice whether or how many groups will be found. Cluster analysis is described in more detail in Section 4.8.

4.2 The Concept and Need for Principal Components Analysis

PCA is probably the most widespread multivariate chemometric technique, and because of the importance of multivariate measurements in chemistry, it is regarded, by many, as the technique that most significantly changed the chemist's view of data analysis.

4.2.1 History

There are numerous claims to the first use of PCA in the literature. Probably, the most famous early paper was by Pearson in 1901. However, the fundamental ideas are based on approaches well known to physicists and mathematicians for much longer, namely those of eigenanalysis. In fact, some school mathematics syllabuses teach ideas about matrices that are relevant to modern chemistry. An early description of the method in physics was provided by Cauchy in 1829. It has been claimed that the earliest non-specific reference to PCA in the literature was in 1878, although the author of the paper almost certainly did not realise the potential and was dealing mainly with a problem of linear calibration.

It is generally accepted that the revolution in the use of multivariate methods took place in the 1920s and 1930s. Psychometrics is one of the earliest applications of applied multivariate analysis that developed during this period and one important area involves relating answers in tests to underlying factors, for example, verbal and numerical ability as illustrated in Figure 4.1. PCA relates a data matrix consisting of these answers to a number of psychological 'factors'. In certain areas of statistics, concepts of FA and PCA are intertwined, but in chemometrics, both approaches

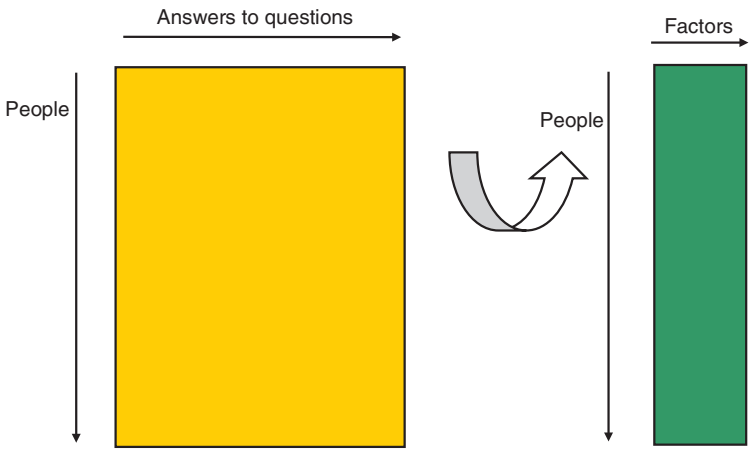


Figure 4.1 Factor analysis in psychology.

have different implications, PCA involves using abstract functions of the data to look at patterns, whereas FA involves obtaining information such as spectra that can be directly related to the chemistry.

Natural scientists of all disciplines, from biologists, geologists and chemists, have caught onto these approaches over the past few decades. Within the chemical community, the first major applications of PCA were reported in the 1970s and form the foundation of many modern chemometric methods described in this and subsequent chapters.

1. Principal components analysis and factor analysis have the same meaning in chemistry.
   (a) True
   (b) False

4.2.2 Multivariate Data Matrices

A key idea is that most chemical measurements are inherently *multivariate*. This means that more than one measurement can be made on a single sample. An obvious example is spectroscopy: we can record a spectrum at hundreds of wavelength on a single sample. Many traditional chemical approaches are *univariate*, in which only one wavelength (or measurement) is used per sample, but this misses much information. Another important application is quantitative structure–property relationships, in which many physical measurements are available on a number of candidate compounds (bond lengths, dipole moments, bond angles, etc.). Can we predict, statistically, the biological activity of a compound? Can this assist in pharmaceutical drug development? Several pieces of information are available. PCA is one of the several multivariate methods that allows us to explore patterns in these data, similar to exploring patterns in psychometric data. Which compounds behave similarly? Which people belong to a similar group? How can this behaviour be predicted from available information?

As an example, consider a metabolomics experiment in which mice of identical genetic makeup (haplotypes) were fed different diets at different periods of time. Extracts of their urine were analysed by NMR. The results can be organised as a matrix, whose rows represent the individual mice (or samples) and the columns the NMR peaks (or variables), as illustrated in Figure 4.2. The individual numbers in the matrix are called *elements* – not to be confused with chemical elements. The data are multivariate because several pieces of information were recorded per sample. If we analysed 200 mouse samples and 1000 NMR peaks, we say that the data are organised into a 200×1000 matrix, with the number of rows cited first.

What might we want to ask about the data? Which mice are similar as judged from their NMR profile of urine extracts? Is diet or feeding time important? Can we see trends according to diet and accordingly how long they were fed? Which NMR signals were markers for diet and can we relate these to specific chemicals? We may have sampled a mouse several times, is the identity of the mouse more important than the diet it was fed? These questions are not about grouping the mice into different classes, but about answering questions whether there are actually any interesting trends, after which we may then want to go on and perform more detailed models or additional experiments. The information about the mice (or their treatment) is mixed up, and it is unlikely that any single NMR peak on its own would be able to answer all our questions. We want a visual or exploratory method to try to convert the huge amount of NMR data into something we can visualise.

Of course, sometimes, the trends can be related to a specific chemical species, for example, in coupled chromatography or equilibrium studies. PCA can still be used as an important exploratory technique, but the principal components, rather than primarily being used for visualisation, can be related to directly interpretable chemical trends. In this chapter, we illustrate the techniques using both types of case studies. In general terms for the former type of data,

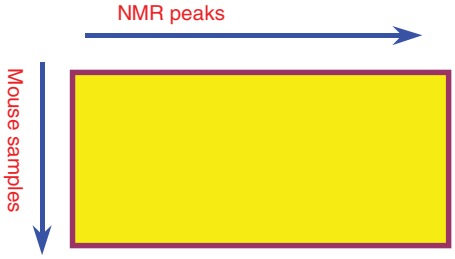


Figure 4.2 Matrix representation of results from a metabolomics experiment.

we may then go on to use the more statistical methods discussed in Chapter 5, whereas for the latter type of data, we may be able to get physically interpretable information as discussed in Chapter 7.

1. An experiment is used to obtain 300 samples of serum extracts from patients both with a disease of varying degrees and controls and measure 30 GCMS peaks. The data can be organised as a matrix of dimensions as follows.
   (a) 300×30
   (b) 30×300

4.2.3 Case Studies

In order to illustrate the main concepts of PCA, we will introduce three case studies.

4.2.3.1 Case Study 1: Resolution of Overlapping Chromatographic Peaks

The first case study is from coupled chromatography. It is not necessary to understand the detailed chemical motivations behind the chromatographic technique. This case study represents the intensity of spectra recorded at a number of wavelengths, which are also sequentially related to time. In fact, many of the earliest applications of PCA in chemistry in the 1960s and 1970s were to overlapping analytical peaks such as in chromatography.

The case study involves a high-performance liquid chromatography–diode array detector (HPLC–DAD) chromatogram sampled at $i = 30$ points in time (each at 1 s intervals) and $j = 28$ wavelengths of approximately 4.8 nm intervals, as presented in Table 4.1 (note that the wavelengths are rounded to the nearest nanometer for simplicity, but the original data were not collected at exact nanometer intervals). Absorbances are presented in absorbance units (AU). For readers not familiar with this application, the data set can be considered to consist of a series of 30 spectra recorded sequentially in time, arising from a mixture of compounds each of which has its own characteristic underlying unimodal time profile (often called an *elution profile*).

The data can be represented by a 30×28 matrix, the rows corresponding to elution times and the columns corresponding to wavelengths. Calling this matrix $\boldsymbol{X}$, and each element x_{ij}, the profile chromatogram at time i

$$X_i = \sum_{j=1}^{28} x_{ij}$$

is given in Figure 4.3 and consists of at least two co-eluting peaks.

1. A diode array chromatogram is recorded at 5 s intervals over 5 min, starting at 5 s, and at 5 nm intervals between 200 and 300 nm. It can be represented by a multivariate matrix of dimensions
   (a) 60×20
   (b) 20×60
   (c) 60×21
   (d) 21×60

4.2.3.2 Case Study 2: Near Infrared Spectroscopy of Food

Near infrared (NIR) spectroscopy is one of the first techniques to use chemometrics, its use dating back from the 1970s. It has an important role in food, pharmaceuticals and petrochemicals. In NIR, it is often not possible to examine the contribution of individual compounds, but it is primarily used to study complex mixtures. Many substances such as foods contain a large number of chemicals: the interest is usually not so much to determine the relative amounts of each in a sample but to study ensemble properties, for example, the provenance of a food stuff or whether a pharmaceutical is within acceptable limits to be marketed or the quality of a fuel. Usually, NIR spectra have first to be transformed to take into account problems such as baseline effects, and in the example we use, this has already been done. Often, NIR spectra show very small differences that are not easily detectable to the eye, which are diagnostic of different origins in a set of samples.

For simplification, we choose an example in which we have reduced the digital resolution to 19 nm, so that there are only 32 wavelengths, and the data have already been pre-processed so that baseline effects have been removed. There

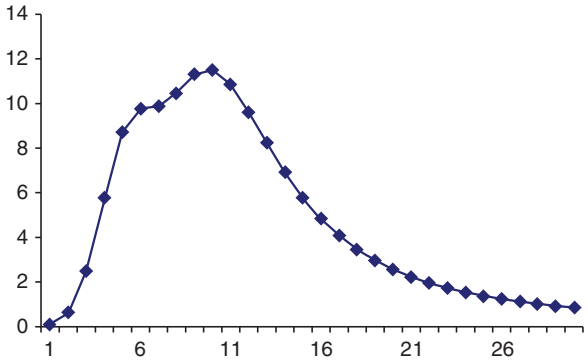


Figure 4.3 Case study 1: chromatographic peak profiles, involving summing intensities of the data from Table 4.1 over all wavelengths.

are numerous quite specialist approaches for preparing NIR data, which are out of the scope of this book, to which the reader is referred to the literature. For illustrative purposes, we are primarily interested in multivariate analysis of the data. The data consist of 72 spectra of edible oils, from four origins, 18 of corn oils, 30 of olive oils, 16 of safflower oils and 8 of corn margarines. The numerical data are presented in Table 4.2, presented as a matrix of 72 rows and 32 columns: each row represents a spectrum of an extract from one of the four groups and each column a wavelength. The elements of the matrix are intensities in AU.

The spectra are presented in Figure 4.4. From this, it can be seen that there are only very small differences between the 72 spectra. The aim of chemometrics is to determine the origins of a sample from its spectrum, for example, which type of oil it belongs to, in this case. By viewing a single spectrum, it is not obvious, but due to the relative stability of NIR spectra (after instrumental artefacts are removed), we will see that using chemometrics, the different groups can be distinguished easily, despite a tiny spectral difference.

1. A series of 50 NIR spectra are recorded, between 800 and 1600 nm inclusive at 10 nm intervals. They can be represented as a data matrix of dimensions
   - **(a)** 50×81
   - **(b)** 50×80
   - **(c)** 81×50
   - **(d)** 80×50

4.2.3.3 Case Study 3: Properties of the Elements

In case studies 1 and 2, the columns (or variables) are sequentially related, as they represent successive wavelengths, and in case study 1, the rows are also sequentially related, representing successive elution times. There is, of course, no requirement of any sequential relationship between successive columns and variables.

Case study 3 is presented in Table 4.3, which consists of the five properties of 27 elements. The second column (group) is not a property but a descriptor. Can we use these five properties to show how elements are related in their chemical reactivity? This 27×5 data matrix is quite different from the ones above. The variables represent quite different things on different scales and thus, as discussed below, have to be handled a bit differently to the previous examples where all measurements are in the same units (AUs). However, in common with the other examples, it is our desire to end up with a graphical simplification of the data to see what are the main trends.

In order to handle this, the data have to be converted to a similar scale, and this is done by standardisation, a technique described later in Section 4.6.4.

1. For a data matrix to be handled by PCA
   - **(a)** All measurements in a row must be in the same physical unit, but this is not a requirement for a column.
   - **(b)** All measurements in a column must be in the same physical unit, but this is not a requirement for a row.
   - **(c)** All measurements in the data matrix must be in the same physical unit.

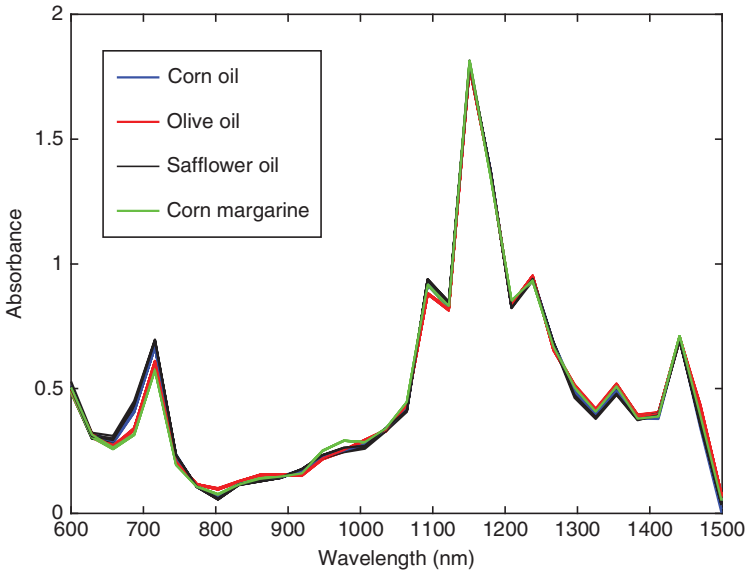


Figure 4.4 Case study 2: superimposed NIR spectra corresponding to the data in Table 4.2.

4.2.4 Aims of PCA

There are two principal needs in chemistry. The first is when PCs can be related to physically interpretable, underlying processes or compounds; thus, the number or quantitative values of the PCs can be directly interpreted in terms of the underlying chemistry. The second is when PCA is primarily used in a qualitative way for visualisation and simplification of data.

In the case of the example of case study 1, we would like to extract information from the two-way chromatogram, which relates to a discrete number of co-eluting compounds in the chromatographic mixture.

- The number of significant PCs is ideally equal to the number of significant compounds. If there are three compounds in the mixture, then we expect that there are only three significant PCs.
- Each PC is characterised by two pieces of information, the *scores*, which, in the case of chromatography, relate to the elution profiles, and the *loadings*, which relate to the spectra.

Below we will discuss in more detail how this information is obtained. However, the ultimate information is physically interpretable.

Case studies 2 and 3 primarily use PCA in a descriptive way. We are, for example, interested in whether the NIR spectra can be used to distinguish four groups of samples and, if so, what wavelengths are characteristic of each group. We are also interested in whether the elements can be distinguished from their physical properties. PCA is used to convert a table of numbers into something pictorial. Of course, where a sample fits into a graph obtained by PCA can also tell us a lot about its provenance.

1. Principal components analysis is applied to the identified GCMS peaks of extracts from 100 patients, 50 with a disease and 50 controls without the disease.
   (a) PCA is primarily used to determine a quantitative model as to whether patients have a disease or not
   (b) PCA is primarily used to determine a qualitative picture of whether the two groups can be separated, and if so which peaks are likely to be responsible

4.3 Principal Components Analysis: The Method

4.3.1 Scores and Loadings

PCA involves an abstract mathematical transformation of the original data matrix, usually obtained experimentally, but sometimes transformed mathematically first, as discussed in Section 4.6, which takes the form

$$\boldsymbol{X} = \boldsymbol{TP} + \boldsymbol{E}$$

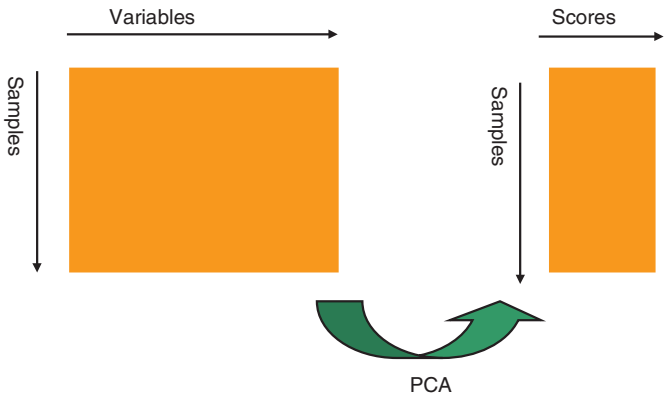


Figure 4.6 PCA as a form of variable reduction.

which is the non-matrix version of the fundamental PCA equation above. Hence, the estimated value of x for the data in Table 4.1 at the 10th wavelength (263 nm) and 8th point in time (true value of 0.304) is given by

- 2.334*0.121 = 0.282 for a one-component model and
- 2.334*0.121 + 0.180*0.123 = 0.304 for a two-component model,

suggesting that two PCs do provide a good estimate of the intensity at this wavelength and point in time.

1. Fifty samples of plasma extracts are recorded spectroscopically using 150 Raman wavelengths. Four principal components are calculated. Using the definitions in this section
   - **(a)** The scores matrix has dimensions 50×4 and the loadings matrix 150×4.
   - **(b)** The scores matrix has dimensions 50×4 and the loadings matrix 4×150.
   - **(c)** The scores matrix has dimensions 4×50 and the loadings matrix 4×150.
   - **(d)** The scores matrix has dimensions 4×50 and the loadings matrix 150×4.
2. Using the NIPALS algorithm
   - **(a)** Scores are normalised but loadings are not.
   - **(b)** Scores are not normalised but loadings are.
   - **(c)** Both scores and loadings are normalised.
3. If data are uncentred
   - **(a)** Scores of different PCs are orthogonal and uncorrelated.
   - **(b)** Scores of different PCs are neither orthogonal nor uncorrelated.
   - **(c)** Scores of different PCs are orthogonal but usually have some correlation.

4.3.2 Rank and Eigenvalues

A fundamental next step is to determine the number of significant PCs that can best characterise a matrix. In a series of mixture spectra or portion of a chromatogram, this should, ideally, correspond to the number of compounds under observation, and there has been a great deal of chemometric literature on methods for determining how many PCs are best for modelling a data set.

The *rank* of a matrix is a mathematical concept that equals to the number of components in a data set and is the number of PCs that exactly model a matrix, so that $\boldsymbol{E}$ is zero, and is usually equal to the smaller of the number of samples or variables, unless there are correlations.

However, in chemistry, we often talk of the approximate or chemical rank of a matrix. This represents the number of components deemed significant, for example, so that the error matrix is relatively very small. There is a huge literature here, and it is important to find how many components are significant, they may, for example, represent the number of significant compounds in a mixture. We may have recorded 50 spectra over 100 wavelengths, but there are only six

compounds. It can also be used to simplify data and remove noise. The chemical rank is usually regarded as the number of significant PCs.

When using PCA for EDA, the number of significant PCs does not usually even have to correspond to the number of compounds in a mixture (if appropriate). For example, if we consider case study 2 (the NIR of a samples of oils), there may be hundreds of detectable compounds, but we are not primarily interested in this: we are interested in whether we can distinguish these groups. The number of significant PCs is an indication of how much we can reduce the data, for example, from 32 wavelengths to 2 or 3 new 'factors' or PCs that are sufficient to describe the data. The aim is to find A so that $\boldsymbol{X} \approx \boldsymbol{TP}$. The matrix $\boldsymbol{E}$ is considered noise.

4.3.2.1 Eigenvalues

Normally after PCA, the *size* of each component can be measured. This is often called an *eigenvalue*: the earlier (and more significant) the components, the larger their size. There are a number of definitions in the literature, but a simple one defines the eigenvalue of a PC as the sum of squares of the scores, so that

$$g_a = \sum_{i=1}^{I} t_{ia}^2$$

where g_a is the ath eigenvalue.

The *sum* of all non-zero eigenvalues for a data matrix equals the sum of squares of the entire data matrix so that

$$\sum_{a=1}^{K} g_a = \sum_{i=1}^{I} \sum_{j=1}^{J} x_{ij}^2$$

where K is the smaller of I or J. Note that if the data are pre-processed before PCA, as discussed in Section 4.6, x must likewise be pre-processed for this property to hold.

Frequently, eigenvalues are presented as percentages, for example, of the sum of squares of the entire (pre-processed) data set, or

$$V_a = 100 \frac{g_a}{\sum_{i=1}^{I} \sum_{j=1}^{J} x_{ij}^2}$$

Successive eigenvalues correspond to smaller percentages.

The *cumulative* percentage eigenvalue is often used to determine (approximately) the proportion of the data modelled using PCA and is given by $\sum_{a=1}^{A} g_a$. The closer this the total sum of squares of the (pre-processed) data, the more faithful the model. The percentage can be plotted against the number of eigenvalues in the PC model.

It is an interesting feature that the *residual sum of squares* (RSSs)

$$\mathrm{RSS}_A = \sum_{i=1}^{I} \sum_{j=1}^{J} x_{ij}^2 - \sum_{a=1}^{A} g_a$$

after A eigenvalues also equals the sum of squares for the error matrix, between the PC model and the raw data, whose elements are defined by

$$e_{ij} = x_{ij} - \hat{x}_{ij} = x_{ij} - \sum_{a=1}^{A} t_{ia} p_{aj}$$

or

$$\sum_{i=1}^{I} \sum_{j=1}^{J} x_{ij}^2 - \sum_{a=1}^{A} g_a = \sum_{i=1}^{I} \sum_{j=1}^{J} e_{ij}^2$$

because the product of the elements of any two different eigenvectors is 0, as discussed in Section 4.3.1.

The first three eigenvalues for the data in Table 4.1 are presented in Table 4.5. The total sum of squares of the entire data set is 61.00, allowing the various percentages to be calculated. It can be seen that two eigenvalues represent more than 99.99% of the data in this case. In fact, the interpretation of the size of an eigenvalue depends, in part, on the

However,

- the sum of squares of the original data $\boldsymbol{X}$ equals 135, as the data have been standardised and there are 27×5 measurements (for standardised data, the total sum of squares equals the number of elements in the data, as discussed in Section 4.6.4).

Therefore,

- the sum of squares of the error matrix $\boldsymbol{E}$ equals $135 - 129.94$ or 5.06: this number is also equal to the sum of eigenvalues 4 and 5, as there are only two non-zero eigenvalues in the data.

Sometimes, the eigenvalues can be interpreted in physical terms. For example,

- the data set in Table 4.1 consists of 30 spectra recorded in time at 28 wavelengths,
- the error matrix is of size 30×28, consisting of 840 elements,
- but the error sum of squares after $a = 1$ PC equals $60.99 - 59.21 = 1.78$ AU^2,
- so the root mean square error is equal to $\sqrt{1.78/840} = 0.046$ (in fact, some chemometricians adjust this for the loss of degrees of freedom due to the calculation of one PC but because 840 is a large number, this adjustment is small and we will stick to the convention in this book of dividing x errors simply by the number of elements in the data matrix).

Is this a physically sensible number? This depends on the original units of measurement and what the instrumental noise characteristics are. If it is known that the root mean square noise is about 0.05 units, then it seems sensible. If the noise level, however, is substantially lower, then not enough PCs have been calculated. In fact, most modern chromatographic instruments can determine peak intensities much more accurately than 0.05 AU; hence, this would suggest that a second PC is required.

The principle of examining the size of successive eigenvalues can be extended, and in spectroscopy, a large number of so-called *indicator* functions have been proposed, many by Malinowski, whose text on FA is a classic. Most functions involve producing graphs of functions of eigenvalues and predicting the number of significant components using various criteria. Over the past decade, several new functions have been proposed, some based on distributions such as the F-test. For more statistical applications such as quantitative structure–activity relationships (QSAR), these indicator functions are not so applicable, but in spectroscopy and certain forms of chromatography where there are normally a physically defined number of factors and well-understood error (or noise) distributions, such approaches are valuable. From a simple rule of thumb, knowing (or estimating) the noise distribution, for example, from a portion of a chromatogram or spectrum where there is known to be no compounds (or a blank), and then determining how many eigenvalues are required to reduce the level of error, allows an estimate of significant components.

1. The total sum of squares of a data matrix is equal to 735.21, and its first eigenvalue is equal to 512.60 and its second to 48.91. Therefore, the sum of squares of the remaining error matrix is
   (a) 173.70
   (b) 48.91
   (c) 22.61
2. A series of spectra are recorded for50 samples at 32 wavelengths. The data are standardised and the sum of the first three eigenvalues is 1257. The per cent of the data represented by the first three eigenvalues is
   (a) 88.6%
   (b) 78.6%

4.3.2.2 Cross-validation

A complementary series of methods for determining the number of significant components is based on cross-validation. It is assumed that significant components model 'data', whereas later (and redundant) components model 'noise'. Auto-predictive models involve fitting PCs to the entire data set and always provide a closer fit to the data the more the components are calculated. Hence, the residual error will be smaller if 10 rather than nine PCs are calculated. This does not necessarily indicate that it is correct to retain all 10 PCs; the later PCs may model noise that we do not want.

As in the case of most chemometric methods, there are innumerable variations on the theme, and it is important to be careful to check every author in detail. However, the LOO method described above is popular and relatively easy to implement and understand.

1. Comparing PRESS and RSS which is true.
   (a) For the same number of PCs, PRESS is likely to be larger than RSS.
   (b) For the same number of PCs, PRESS is likely to be less than RSS.
   (c) There is not much relationship between the two.

4.4 Factor Analysis

Principal components are primarily abstract mathematical entities with certain well-defined properties such as orthogonality. Chemists, however, often like to think in terms of physically interpretable components, such as spectra or concentration profiles, and these are often called *factors*. Note that there is a difference between the chemist's and statistician's terminology: many statisticians would call principal components factors instead.

As an illustration, we will use the case of coupled chromatography, such as HPLC–DAD, as in case study 1. For a simple chromatogram, the underlying data set can be described as a sum of responses for each significant compound in the data, which are characterised by (a) an elution profile and (b) a spectrum, plus noise or instrumental error. In matrix terms, this can be written as

$$\boldsymbol{X} = \boldsymbol{CS} + \boldsymbol{E}$$

where

- $\boldsymbol{X}$ is the original data matrix or coupled chromatogram,
- $\boldsymbol{C}$ is a matrix consisting of the elution profiles of each compound,
- $\boldsymbol{S}$ is a matrix consisting of the spectra of each compound and
- $\boldsymbol{E}$ is an error matrix.

Consider the matrix of case study 1, a portion of a chromatogram recorded over 30 points in time and 28 wavelengths, consisting of two partially overlapping compounds.

- $\boldsymbol{X}$ is a matrix of 30 rows and 28 columns,
- $\boldsymbol{C}$ is a matrix of 30 rows and two columns, each column corresponding to the elution profile of a single compound,
- $\boldsymbol{S}$ is a matrix of two rows and 28 columns, each row corresponding to the spectrum of a single compound,
- $\boldsymbol{E}$ is a matrix of the same size as $\boldsymbol{X}$.

If we observe $\boldsymbol{X}$, can we then predict $\boldsymbol{C}$ and $\boldsymbol{S}$? In previous chapters, we have used a 'hat' notation to indicate a *prediction*, so it is also possible to write the above equation as

$$\boldsymbol{X} \approx \hat{\boldsymbol{C}}\hat{\boldsymbol{S}}$$

Ideally, the predicted spectra and chromatographic elution profiles are close to the true ones, but it is important to realise that we can *never directly or perfectly* observe the underlying data. There will always be measurement error even in practical spectroscopy. Chromatographic peaks may be partially overlapping or even embedded, which means that chemometric methods will help resolve the chromatogram into individual components.

One aim of chemometrics is to obtain these predictions after first treating the chromatogram as a multivariate data matrix and then performing PCA. Each compound in the mixture is a 'chemical' factor with its associated spectra and elution profile, which can be related to principal components, or 'abstract' factors, by a mathematical transformation.

It is possible to relate PCs to chemical information, such as, for example, elution profiles and spectra in diode array HPLC by

$$\hat{\boldsymbol{X}} = \boldsymbol{TP} = \hat{\boldsymbol{C}}\hat{\boldsymbol{S}}$$

The conversion from 'abstract' to 'chemical' factors is sometimes called a *rotation* or *transformation* and examples will be discussed in more detail in Chapter 7, briefly it involves finding a square matrix $\boldsymbol{R}$ so that

$$\boldsymbol{TP} = \boldsymbol{TRR}^{-1}\boldsymbol{P} = \hat{\boldsymbol{C}}\hat{\boldsymbol{S}} \quad \text{or} \quad \hat{\boldsymbol{C}} = \boldsymbol{TR} \quad \text{and} \quad \hat{\boldsymbol{S}} = \boldsymbol{R}^{-1}\boldsymbol{P}$$

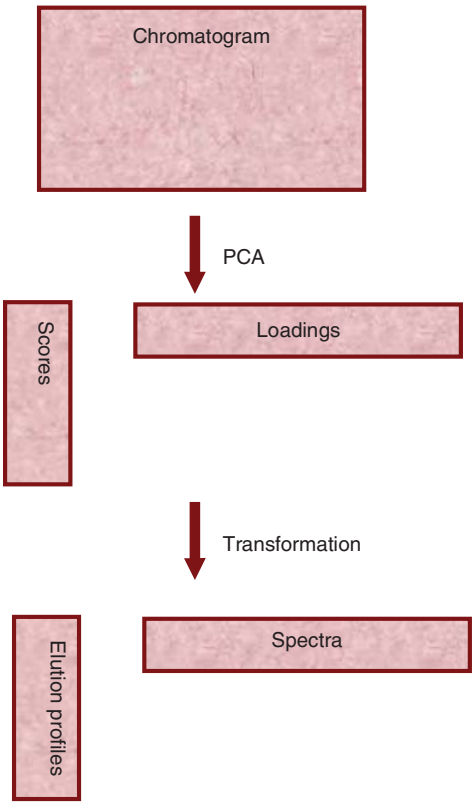


Figure 4.8 Relationship between PCA and factor analysis in coupled chromatography.

and is illustrated in Figure 4.8. Note that FA in chemometrics is by no means restricted to chromatography. An example is the pH titration profile of a number of species containing a different number of protons together with their spectra. Each equilibrium species has a pH titration profile and a characteristic spectrum. We will discuss other applications in Chapter 7.

FA is often called by a number of alternative names such as 'rotation' or 'transformation' but is a procedure used to relate the abstract PCs to meaningful chemical factors; the influence of Malinowski in the 1980s has introduced this terminology into chemometrics.

It is important to understand that the historic development of multivariate methods in chemometrics came from two very different perspectives. The first arose from physical chemistry and quantitative analytical chemistry. Multivariate methods were used to estimate physically interpretable things such as reaction rates of spectral profiles. The other came from statistics and PCs were primarily viewed as abstract entities helped to simplify and interpret data. The terminology of FA in chemistry mainly came from the former group.

1. $\boldsymbol{R}$ is used in the transformation from $\boldsymbol{T}$ (an abstract factor) to $\hat{\boldsymbol{C}}$, where there are 30 samples and three principal components
   - **(a)** $\boldsymbol{R}$ is a 3×3 matrix
   - **(b)** $\boldsymbol{R}$ is a 30×3 matrix
   - **(c)** $\boldsymbol{R}$ is a 30×30 matrix

4.5 Graphical Representation of Scores and Loadings

Many revolutions in chemistry relate to the graphical presentation of information. For example, fundamental to the modern chemist's way of thinking is the ability to draw structures on paper in a convenient and meaningful manner. Years of debate preceded the general acceptance of the Kekulé structure for benzene: today's organic chemist can write

down and understand complex structures of natural products without the need to plough through pages of numbers of orbital densities and bond lengths. Yet, underlying these representations are quantum mechanical probabilities; hence, the ability to convert from numbers to a simple diagram has allowed a large community to clearly think about chemical reactions.

Hence, with statistical data, and modern computers, it is easy to convert from numbers to graphs. Many modern multivariate chemometricians think geometrically as much as numerically, and concepts such as principal components are often treated as much as objects in an imaginary multivariate space than mathematical entities. The algebra of multi-dimensional space is the same as that of multivariate statistics. Older texts, of course, were written before the days of modern computing; hence, the ability to produce graphs was more limited. However, it is now possible to obtain a large number of graphs rapidly using simple software, much is even possible using Excel. There are many ways of visualising PCs. Below we will primarily look at graphs of the first two PCs, for simplicity.

4.5.1 Scores Plots

One of the simplest plots is that of the scores of one PC against the other. Figure 4.9 illustrates the PC plot of the first two PCs against each other obtained from case study 2, corresponding to plotting a graph of the first two columns of the matrix $\boldsymbol{T}$. The horizontal axis is the scores for the first PC and the vertical axis for the second PC. No pre-processing has been performed. Note that the NIPALS algorithm has been used: this affects the scale of the axes; if SVD were used, the axes would be normalised; thus, when employing different packages, always check what algorithm has been employed. In this text, we restrict to NIPALS for simplicity.

From this 'picture', we can clearly see that the four groups are distinguished by the NIR spectra. Note that in the PCA algorithm, the sign of the PC cannot be controlled; hence, some packages may invert the axes because the sign of a square root can be either negative or positive.

It is possible to present the data in three dimensions, in this case adding the scores of PC3 as the third axis, as shown in Figure 4.10. Although these graphical representations look nice, in fact, the third PC adds little to the separation. The scores can also be presented in one dimension, as shown in Figure 4.11. We can see that all groups are in fact separated using PC1, but the distinction between safflower oil and corn oil is more evident using PC2; however, corn margarine and olive oil are not separable in PC2. If we had not coloured the groups, we may not have easily been able to distinguish all four using a single component, but once both are plotted, we can see all groups are easily distinguishable.

Case study 1 has different characteristics to case study 2. In this case, the observations are relating sequentially in time, whereas in case study 1, there is no specific sequence to the data.

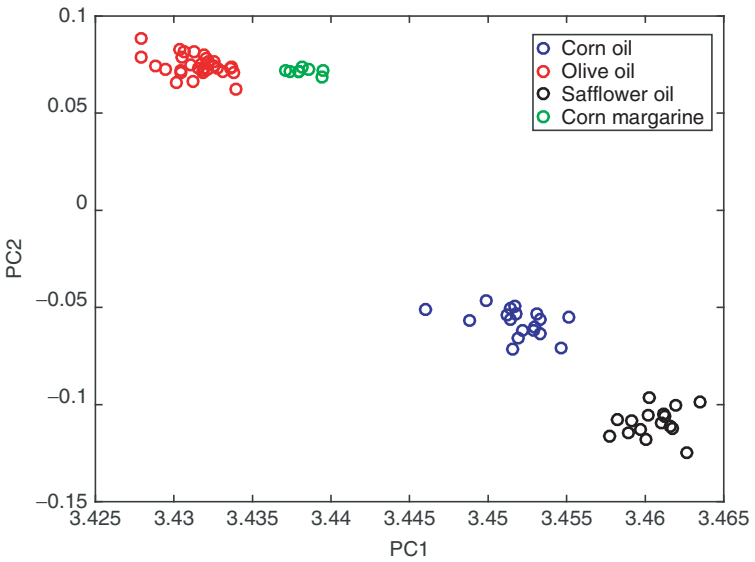


Figure 4.9 Plot of scores of PC2 versus PC1 for case study 2.

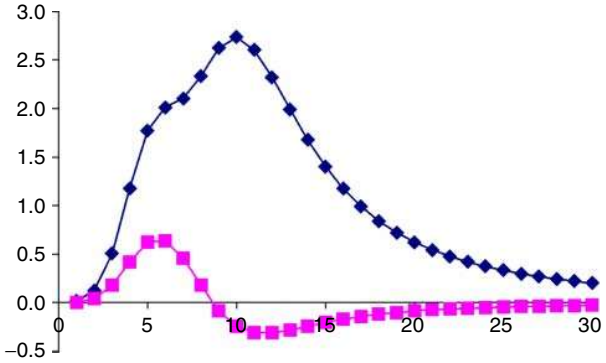


Figure 4.13 Scores of the first two PCs of case study 1 versus sample number.

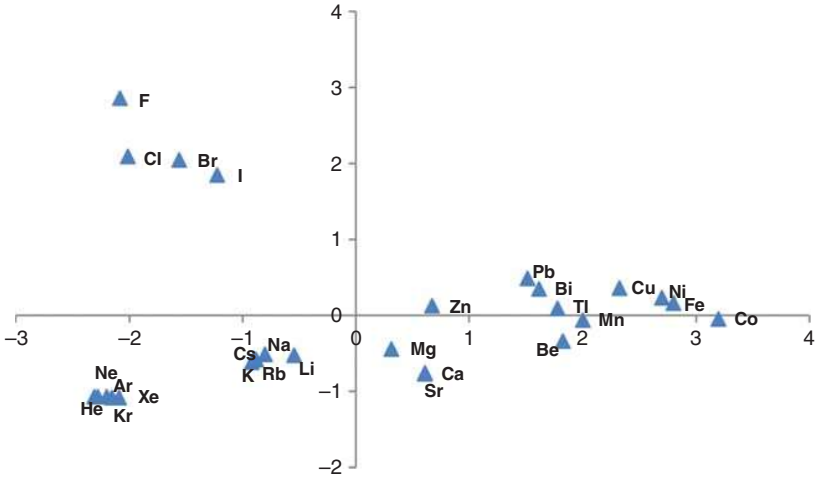


Figure 4.14 Scores of principal component 2 (vertical axis) versus principal component 1 (horizontal axis) for the standardised data of case study 3.

1. Four PCs are calculated. How many 2D dimensional graphs of the scores of one PC against another can be obtained?
   (a) 3
   (b) 4
   (c) 6
2. A chemical process is studying by spectroscopy as a function of pH. A graph of the scores of the first few components as a function of pH is likely to provide us with insights into the change in chemical species with pH.
   (a) True
   (b) False

4.5.2 Loadings Plots

It is not, however, only the scores that are of interest but also the loadings. Exactly, the same principles apply; for example, the value of the loadings at one PC can be plotted against that at the other PC.

The result for the first two PCs for case study 1 is shown in Figure 4.15. This figure looks quite complicated because both spectra overlap and absorb at similar wavelengths. The pure spectra are presented in Figure 4.16. Now we can understand a little more about these graphs. From this, we can see that the top right-hand corner of the scores plot corresponds to a direction for the fastest eluting compound (=**A**), whereas the bottom right-hand corner to the slowest eluting compound (=**B**). Similar interpretation can be obtained from the loadings plots. Wavelengths in the bottom

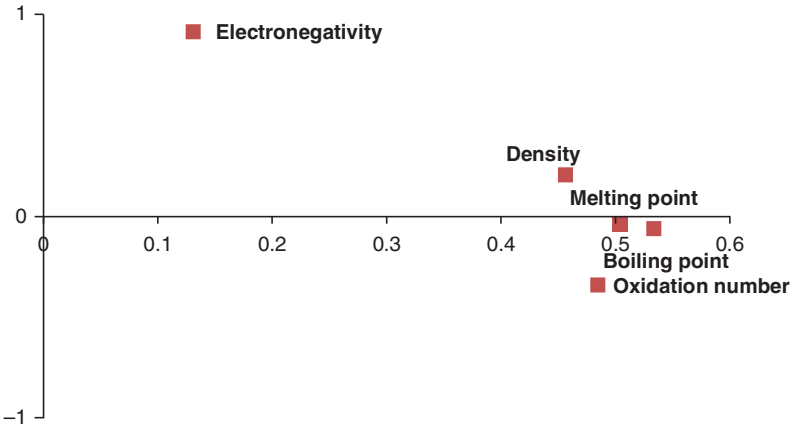


Figure 4.19 Loadings of principal component 2 versus principal component 1 for the standardised data of case study 3.

(which compounds have loadings corresponding to a specific group of samples), to look for diagnostic wavelengths or chromatographic peaks.

1. Loadings plots can be compared with scores plots to suggest diagnostic variables.
   (a) True
   (b) False
2. Three loadings are calculated for a data set. It is possible to obtain
   (a) One 3D loadings plot and two 2D loadings plots of different PCs against each other.
   (b) One 3D loadings plot and three 2D loadings plots of different PCs against each other.
   (c) One 3D loadings plot and one 2D loadings plot of different PCs against each other.
   (d) Three 3D loadings plots and two 2D loadings plots of different PCs against each other.

4.6 Pre-processing

All chemometric methods are influenced by the method for data pre-processing, or preparing information before application of mathematical algorithms. An understanding is essential for correct interpretation from multivariate data packages, but will be illustrated with reference to PCA, and is one of the first steps in data preparation. It is often called *scaling* or *data transformation* and the most appropriate choice can relate to the chemical or physical aim of the analysis. Pre-processing is normally performed before PCA, but in this chapter, it is introduced afterwards as it is hard to understand how it influences the resultant models without first appreciating the main concepts of PCA.

4.6.1 Transforming Individual Elements of a Matrix

One of the first considerations is whether individual elements of a matrix need transformation. The measurement units may mask the trends we are looking for. As an example, we often measure acidity or alkalinity in units of pH rather than $[H^+]$; hence, it is sometimes useful to transform raw measurements For example, we may be interested in the distribution of biomarkers using chromatographic peak intensities. Sometimes, minor metabolites are of low concentrations, but their relative variation may be as interesting as the major ones.

Logarithmic scaling is a common means of performing this transformation, where x_{ij} is replaced by $\log(x_{ij})$ either to the base 10 or as natural logs. The difficulty here is where some readings are recorded by zero: this is quite common if measurements are below the limit of detection, a logarithm of zero is undefined. Under such circumstances, these zero values are usually replaced by a small positive number, for example, half the lowest detected value for the variable in question, or a small baseline is added to every number. If there are too many zeroes in a data set, logarithmic scaling

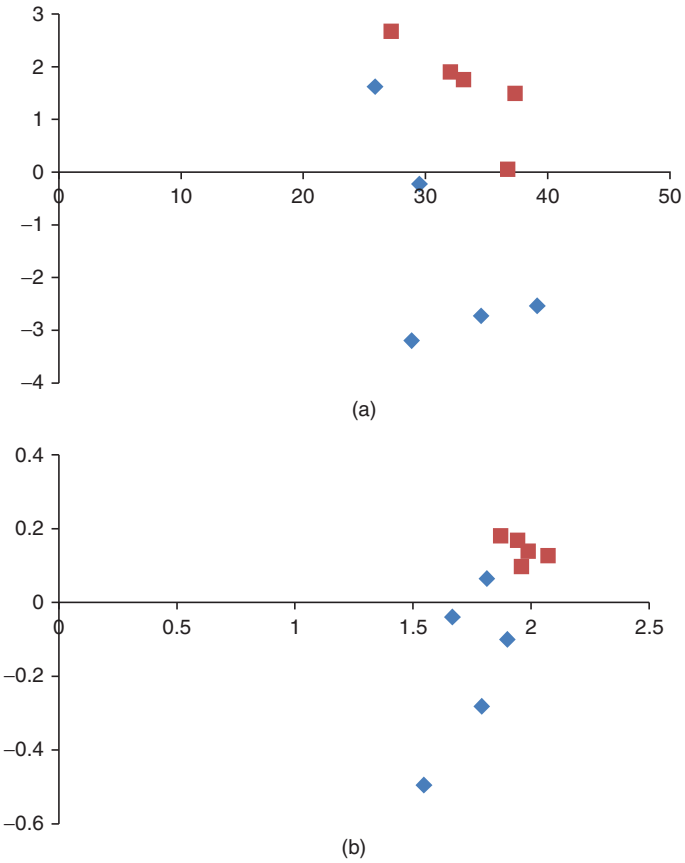


Figure 4.20 Scores of the first two PCs of the data in Table 4.11, (a) raw data (b) log scaled data.

4.6.2 Row Scaling

Another common problem is when the amount of a sample is hard to control. For example, it may be hard to obtain an identical amount of an extract. Under such circumstances, we are primarily interested in proportions rather than absolute amounts. We may be interested in a rock sample and the proportion of different elements found, but each sample may be of very different size. This problem is sometimes called one of closure and in fact can introduce some quite specific problems into the data.

However, it is usual under these circumstances to scale the rows (i.e. all the measurements on each sample) to a specific total. The simplest is to scale each row to a constant total so that x_{ij} is replaced by $x_{ij} / \sum_{j=1}^{J} x_{ij}$. The sum over all measured variables then becomes 1. There are other variants; for example, if there is a standard (internal or external), all peaks or concentrations are referenced to this standard, which may, for example, be an added compound of known concentration. Row scaling usually assumes that all values are positive. If the data have been first transformed using logarithms, great care should be taken, and if such a transformation is felt necessary, alternatives such as replacing x_{ij} by $\sqrt{x_{ij}}$ is normally preferred.

Row scaling to a constant total is sometimes also called *normalisation*, but this term can have many different meanings, so we avoid it. Row scaling can make a substantial difference to the resultant analysis. It is usually only sensible if all variables are of approximately the same intensity. If, for example, one variable was to correspond to the concentration of a compound that is found in high amounts (as variable 4 of the data in Table 4.11), scaling is dominated by this variable. There are various ways of overcoming this, for example, to log scale the data, or ignore some variables or even divide the data into groups. In spectroscopy where wavelengths are all measured on the same scale, row scaling to a constant total usually is quite robust.

An example is presented in Table 4.12 involving a 10×8 matrix (10 samples and eight variables). The rows are scaled to a constant total of 1. The graphs of the scores of PC2 versus PC1 are presented in Figure 4.21. Although the raw

Table 4.12 Example for row scaling.

(a) Raw data							
8.47	11.20	14.67	18.57	20.51	17.98	12.03	6.18
0.34	0.43	0.53	0.65	0.73	0.66	0.48	0.27
10.41	12.35	14.59	17.43	19.52	18.68	14.48	8.84
2.60	3.00	3.44	4.04	4.54	4.47	3.60	2.28
3.79	4.25	4.74	5.45	6.16	6.23	5.22	3.42
5.15	5.53	5.84	6.45	7.37	7.86	7.04	4.87
8.88	9.26	9.37	10.04	11.56	12.87	12.09	8.65
14.86	14.79	14.00	14.16	16.56	19.87	20.07	15.08
13.02	12.72	11.70	11.51	13.56	16.83	17.53	13.42
7.00	6.71	5.97	5.69	6.77	8.74	9.40	7.33
(b) Row scaled data							
0.077	0.102	0.134	0.169	0.187	0.164	0.110	0.056
0.084	0.104	0.129	0.159	0.177	0.162	0.117	0.067
0.090	0.106	0.125	0.150	0.168	0.161	0.125	0.076
0.093	0.107	0.123	0.144	0.162	0.160	0.129	0.082
0.096	0.108	0.121	0.139	0.157	0.159	0.133	0.087
0.103	0.110	0.116	0.129	0.147	0.157	0.140	0.097
0.107	0.112	0.113	0.121	0.140	0.156	0.146	0.105
0.115	0.114	0.108	0.109	0.128	0.154	0.155	0.117
0.118	0.115	0.106	0.104	0.123	0.153	0.159	0.122
0.121	0.116	0.104	0.099	0.117	0.152	0.163	0.127

data does suggest, when the 10 samples are connected in sequence, that there is some sort of trend, once the data are normalised, this is very clear. In fact, the data were simulated by multiplying a 10×2 $\boldsymbol{C}$ matrix by a 2×8 $\boldsymbol{S}$ matrix, each row of which was multiplied by another randomly generated scalar $\boldsymbol{v}_i$, as illustrated in Table 4.13. Row scaling in this example dramatically reveals the underlying trends. Of course, this does not always happen, there may not necessarily be a sequential meaning to the data, for example; however, it is appropriate in certain circumstances.

Row scaling is normally performed after transformation, as discussed in the previous section, but before any methods for column scaling. Note that it is rarely meaningful to row scale where there are negative values.

1. Row scaling is the same as standardisation.
   (a) True
   (b) False

4.6.3 Mean Centring

It is, also, possible to mean centre the columns by subtracting the mean of each column (or variable) so that to transform x_{ij} to $x_{ij} - \overline{x}_j$.

In most traditional statistics, this is always done before PCA because the majority of statistical methods involve looking at variation around a mean. However, in some areas such as signal analysis, the key is to look at variation above a baseline (e.g. in chromatography and spectroscopy), so this default is not always necessary. We will illustrate this by the data of Table 4.14.

The scores plot of PC2 versus PC1 for the data in Table 4.14 before column centring is shown in Figure 4.22 and after on Figure 4.23. Note that the latter plot is not a simple transformation of Figure 4.22 about the origin. In fact, the process of centring is quite complicated, but in this case, the first PC approximates to the average intensity and by removing this we are looking at the variation around this mean. Hence, PC1 of the mean centred data is quite similar

Table 4.14 Example for Mean Centring.

1.161	1.535	2.010	2.545	2.811	2.464	1.648	0.847
1.240	1.545	1.918	2.359	2.624	2.405	1.741	0.987
1.225	1.453	1.717	2.051	2.297	2.198	1.704	1.040
1.285	1.482	1.700	1.993	2.243	2.205	1.778	1.127
1.344	1.511	1.684	1.936	2.189	2.213	1.852	1.214
1.476	1.587	1.674	1.851	2.113	2.255	2.018	1.396
1.501	1.565	1.584	1.696	1.953	2.176	2.043	1.463
1.551	1.543	1.461	1.477	1.727	2.073	2.094	1.574
1.664	1.626	1.495	1.471	1.733	2.152	2.241	1.716
1.742	1.670	1.487	1.417	1.684	2.175	2.339	1.824

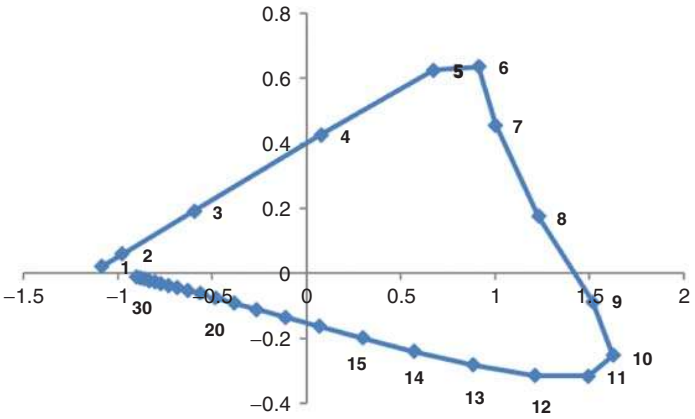


Figure 4.24 Scores plot of PC2 versus PC1 for case study 1 after centring.

Mean centring often has a significant influence on the relative size of the first eigenvalue, which may be reduced dramatically in relative size, and can influence the apparent number of significant components in a data set.

1. Is the prime effect of column mean centring to shift the origin of a PC scores plot?
   (a) Yes always.
   (b) It depends on the case study.
   (c) No.

4.6.4 Standardisation

Standardisation is another common method for data scaling, sometimes called *auto-scaling*, and occurs after mean centring: each variable is also divided by its standard deviation to transform x_{ij} to

$$\frac{x_{ij} - \bar{x}_j}{\sqrt{\sum_{i=1}^{I} (x_{ij} - \bar{x}_j)^2 / I}}$$

The population standard deviation is used rather than the sample standard deviation (see Section A.3.1.2), as this is used as a method of mathematical transformation rather than statistical estimation. Often, variables are on quite different scales, consider case study 3. If they are not standardised, numerically large variables will have a very large influence on the resultant PCs.

1. Standardised data are always mean centred.
   (a) True
   (b) False
2. A data set consists of 30 samples and 12 variables. The sum of squares of the standardised data set is
   (a) 30
   (b) 12
   (c) 360
   (d) 330

4.6.5 Further Methods

There is a very large battery of methods for data pre-processing, although the ones described above are the most common.

- It is possible to combine approaches, for example, first to scale the rows to a constant total and then standardise a data set.
- Weighting of each variable according to any external criterion of importance is sometimes employed. When we standardise a data set, we usually assume that each variable has equal importance.
- Block scaling can be used down the columns analogous to standardisation. If, for example, we record 20 HPLC peaks but 100 NIR wavelengths, we could scale each block so that the total sum of squares is 1 or a defined number, rather than *IJ*. This means that the NIR information does not overwhelm the HPLC information. This also, of course, means that each individual NIR measurement is 1/5 (=20/100) the significance of each HPLC measurement.
- The Box Cox transformation is sometimes used as an alternative to power and logarithmic transformations of individual elements. The aim is to transform each variable into a distribution resembling the normal distribution. For certain statistical tests, the assumption of normality is important.

Undoubtedly, however, the appearance and interpretation not only of PC plots but of almost all chemometric techniques depend on data pre-processing. The influence of pre-processing can be quite dramatic; hence, it is essential for the user of chemometric software to understand and question how and why the data have been scaled or transformed before interpreting the result from a package. More consequences are described in Chapter 7.

1. A data set consists of 50 samples, recorded at 80 UV wavelengths and 120 NIR wavelengths. It is desired to make the NIR and UV information blocks of equal significance.
   (a) Each UV measurement is weighted as being of 1.5 the importance of each NIR measurement.
   (b) Each UV measurement is weighted as being 2/3 the importance of each NIR measurement.
   (c) The individual measurements are standardised for each of the variables.

4.7 Comparing Multivariate Patterns

PC plots are often introduced only by reference to the independent loadings or scores plot of a single data set. Yet, there are common patterns within these different graphs. Consider taking measurements of the concentration of a mineral in a geochemical deposit. This information could be presented as a table of sampling sites and observed concentrations. However, a much more informative approach would be to produce a picture in which physical location and mineral concentration are superimposed, such as a coloured map, each different colour corresponding to a concentration range of the mineral. Two pieces of information are connected, namely geography and concentration. Hence, in many applications of multivariate analysis, one aim may be to connect the samples (e.g. geographical location/sampling site) represented by scores to the variables (e.g. chemical measurements) represented by loadings. Graphically, this requires the superimposition of two types of information.

Another common need is to compare two independent types of measurements. Consider recording the result of a taste panel for a type of food. Their scores relate to the underlying chemical or manufacturing process. A separate

measurement could be chemical, such as a chromatographic or spectroscopic profile. Ideally, the chemical measurements will relate to the taste, can each type of measurement give similar information, and so, can we predict the taste by using analytical chemical techniques?

4.7.1 Biplots

A biplot involves superimposition of a scores and a loadings plot. In order to superimpose each plot on a sensible scale, one approach is to divide the scores to provide scaled scores, which are then superimposed on the loadings as follows:

$$\frac{t_{ia}}{\sum_{i=1}^{I} t_{ia}^2 / I}$$

Notice that if the scores are mean centred, the denominator equals the variance. Some authors use the expression in the denominator of this equation to represent an eigenvalue; hence, in certain articles, it is stated that the scores of each PC are divided by their eigenvalue. As is usual in chemometrics, it is important to recognise that there are many different schools of thought and incompatible definitions.

We will illustrate this by case study 3, as shown in Figure 4.26. At first, the results do not seem clear-cut. For example, electronegativity is at the top but overlapping with the most electronegative elements (the halides). However, this is because the property relates to all the elements. Those at the bottom of the scores plot have low electronegativity and those at the top of the scores plot have high electronegativity. Likewise, other properties such as MP and BP are associated with elements in the right-hand side of the figure, suggesting that these properties are most associated with PC1, a positive value implying high MP and so on, and a negative value low MP. Note that the sign of a PC can differ according to calculation method, but if inverted, both loadings and scores will change in sign, resulting in reflection around the axes.

It is not necessary to restrict biplots to two PCs, but, of course, when more than three are used, graphical representation becomes difficult, and numerical measures of fit between scores and loadings are often employed, using statistical software.

1. A biplot is a plot of one PC against another.
   (a) True
   (b) False

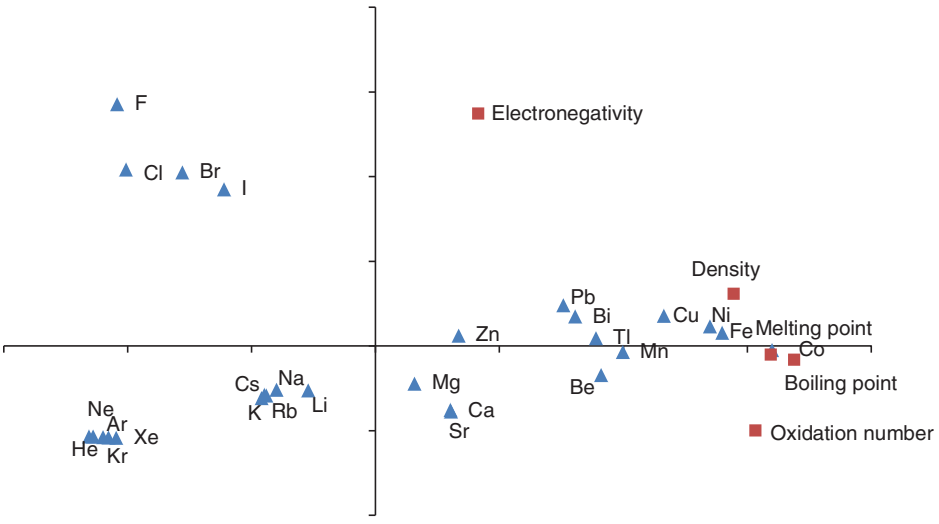


Figure 4.26 Biplot of scores of the first two PCs of case study 3.

4.7.2 Procrustes Analysis

Another important facility is to be able to compare different types of measurements. For example, we may want to determine whether MS and NMR measurements provide similar information. A statistical method called *procrustes analysis* will help us here.

Procrustes was a Greek god who kept a house by the side of the road where he offered hospitality to passing strangers, who were invited for a meal and a night's rest in his very special bed which Procrustes described as having the unique property that its length exactly matched whosoever lay down upon it. What he did not say was the method by which this 'one-size-fits-all' was achieved: as soon as the guest lay down, Procrustes went to work upon them, stretching them if they were too short for the bed or chopping off their legs if they were too long.

Similarly, procrustes analysis in statistics involves comparing two spatial representations of data, such as two PC scores plots. One such plot is the reference and a second plot is manipulated to resemble the reference plot as closely as possible. This manipulation is done mathematically, involving up to four main transformations.

- *Reflection*. This transformation is a consequence of the inability to control the sign of a principal component.
- *Rotation.*
- *Scaling* (*or stretching*). This transformation is because the scales of the two types of measurements may be very different.
- *Translation.*

If two data sets are already standardised, transformation 3 may not be necessary, and the fourth transformation is not often used.

The aim is to reduce the root mean square difference between the scores of the reference data set and the transformed data set

$$r = \sqrt{\sum_{i=1}^{I}\sum_{a=1}^{A}({}^{ref}t_{ia} - {}^{trans}t_{ia})^2 / I}$$

The resultant view gives a consensus of the techniques, but it can also answer how similar two different techniques are.

It is not necessary to restrict each measurement technique to two PCs; indeed, in many practical cases, four or five PCs are employed. Computer software is available to compare scores plots and provide a numeric indicator of the closeness of the fit, but it is not easy to visualise. As PCs do not often have a physical meaning, it is important to recognise that, in some cases, it is necessary to include several PCs for a meaningful result. For example, if two data sets are characterised by four PCs, and each one is of approximately equal size, the first PC for the reference data set may correlate most closely with the third for the comparison data set, thus including only the first two components in the model could result in very misleading conclusions. It is usually a mistake to compare PCs of equivalent significance with each other, especially when their size is fairly similar.

Procrustes analysis can be used to answer quite sophisticated questions. For example, in sensory research, are the results of a taste panel comparable with chemical measurements? If so, can the rather expensive and time-consuming taste panel be replaced by chromatography? A second use of procrustes analysis is to reduce the number of tests: an example being of clinical tests. Sometimes, 50 or more bacteriological tests are performed but can these be reduced to 10 or less? A way to check this is by performing PCA on the results of all 50 tests and compare the scores plot when using a subset of 10 tests. If the two scores plots provide comparable information, the 10 selected tests are just as good as the full set of tests. This can be of significant economic benefit.

1. It is desired to reduce the number of tests used to determine the quality of a fuel. Procrustes analysis can be used for this purpose.
   (a) True
   (b) False

4.8 Unsupervised Pattern Recognition: Cluster Analysis

EDA such as PCA is primarily used to determine general relationships between data. Sometimes, more complex questions need to be answered, such as do the samples fall into groups? Cluster analysis is a well-established approach that

Table 4.16 Example for cluster analysis.

	A	B	C	D	E
1	0.9	0.5	0.2	1.6	1.5
2	0.3	0.2	0.6	0.7	0.1
3	0.7	0.2	0.1	0.9	0.1
4	0.1	0.4	1.1	1.3	0.2
5	1.0	0.7	2.0	2.2	0.4
6	0.3	0.1	0.3	0.5	0.1

was primarily developed by biologists to determine similarities between organisms. Numerical taxonomy emerged from a desire to determine relationships between different species, for example, genera, families and phyla. Many textbooks in biology show how organisms are related using family trees.

The chemist also wishes to relate samples in a similar manner. Can protein sequences from different animals be related and does this tell us about the molecular basis of evolution? Can the chemical fingerprint of wines be related and does this tell us about the origins and taste of a particular wine? Unsupervised pattern recognition employs a number of methods, primarily cluster analysis, to group different samples (or objects) using chemical measurements.

4.8.1 Similarity

The first step is to determine the similarity between objects. Table 4.16 consists of six objects (1–6) and five measurements (A–E). What are the similarities between the objects? Each object has a relationship with the remaining five objects. How can a numerical value of similarity be defined? A similarity matrix can be obtained, in which the similarity between each pair of objects is calculated using a numerical indicator. Notice that it is possible to pre-process data before calculation of a number of these measures (see Section 4.6).

Four of the most popular ways of determining how similar objects are to each other are as follows.

- *Correlation coefficient between samples.* A correlation coefficient of 1 implies that samples have identical characteristics, which all objects have with themselves. Some workers use the square or absolute value of a correlation coefficient, and it depends on the precise physical interpretation as to whether negative correlation coefficients imply similarity or dissimilarity. In this text, we assume that the more negative the correlation coefficient, the less similar the objects; sometimes the square of the correlation coefficient is used instead. The correlation matrix is presented in Table 4.17. Notice that the top right is not presented, as it is the same as the bottom left. The higher the correlation coefficient, the more similar the objects.
- *Euclidean distance.* The distance between the two samples k and l is defined by

$$d_{kl} = \sqrt{\sum_{j=1}^{J} \left(x_{kj} - x_{lj}\right)^2}$$

where there are j measurements, and x_{ij} is the jth measurement on sample i; for example, x_{23} is the third measurement on the second sample, equalling 0.6 in Table 4.16. The smaller this value, the more similar the samples; thus, this

Table 4.17 Correlation matrix.

	1	2	3	4	5	6
1	1.000					
2	−0.041	1.000				
3	0.503	0.490	1.000			
4	−0.018	0.925	0.257	1.000		
5	−0.078	0.999	0.452	0.927	1.000	
6	0.264	0.900	0.799	0.724	0.883	1.000

- *Mahalanobis distance.* This method is popular with many chemometricians, and, while superficially similar to the Euclidean distance, it takes into account that some variables may be correlated and so measure more or less the same properties. The distance between objects k and l is best defined in matrix terms by

$$d_{kl}^2 = (\boldsymbol{x}_k - \boldsymbol{x}_l)\boldsymbol{C}^{-1}(\boldsymbol{x}_k - \boldsymbol{x}_l)'$$

  where $\boldsymbol{C}$ is the variance–covariance matrix of the variables, a matrix symmetric about the diagonal, whose elements represent the covariance between any two variables, of dimensions $J \times J$. See Section A.3.1 for definitions of these parameters; note that one should use the population rather than sample statistics. This measure has a similar computational formula to the Euclidean distance except that the inverse of the variance–covariance matrix is inserted as a scaling factor. However, the computational method discussed above cannot be applied where the number of variables exceeds the number of objects because the variance–covariance matrix would not have an inverse. There are some ways around this (e.g. when calculating spectral similarities where the number of wavelengths far exceeds the number of spectra), such as first performing PCA and then retaining the first few (or all non-zero) PCs or the most significant variables for subsequent analysis. In the case of Table 4.16, the Mahalanobis distance would not be a very useful measure unless either the number of samples is increased or the number of variables are decreased. This distance metric does have very important uses in chemometrics, but more commonly in the areas of supervised pattern recognition, as discussed in Chapter 5, where its properties will be described in more detail. Note in contrast that if the number of variables is very small, although the Mahalanobis distance is an appropriate measure, correlation coefficients are less useful.

There are several other related distance measures in the literature, but normally, quite good reasons are required if a very specialist measure is to be employed.

1. Two samples that are close to each other should have a relatively low distance but high correlation coefficient compared with others in a data set.
   (a) True
   (b) False
2. The Manhattan distance is always ≥ the Euclidean distance ≥ Mahalanobis distance.
   (a) True
   (b) False

4.8.2 Linkage

The next step is to link the objects. The most common approach is called *agglomerative clustering*, in which single objects are gradually connected to each other in groups. Any similarity measure can be used in the first step, but for simplicity, we will illustrate this using only the correlation coefficients presented in Table 4.17. Similar considerations apply to all the similarity measures introduced in Section 4.8.1, except that in the other cases the lower the distance, the more similar the objects, whereas a high correlation represents high similarity in our example.

- From the raw data, find the two objects that are most similar (closest together). According to Table 4.17, these are objects 2 and 5, as they have the highest correlation coefficient (=0.999) (remember that because only five measurements have been recorded, there are only 4 degrees of freedom for the calculation of correlation coefficients, which means quite high values can be obtained fairly easily).
- Next, form a 'group' consisting of these two most similar objects. Four of the original objects (1, 3 and 6) and a group consisting of objects 2 and 5 together remain, leaving a total of five new groups, four consisting of a single original object and one consisting of two 'clustered' objects.
- The tricky bit is to decide how to represent this new grouping. As in the case of distance measures, there are quite a few approaches. The main task is to recalculate the numerical similarity values between the new group and the remaining objects. There are three principal ways of doing this.
  - *Nearest neighbour.* The similarity of the new group from all the other groups is given by the *highest* similarity of either of the original objects to each other object. For example, object 6 has a correlation coefficient of 0.900 with object 2 and 0.883 with object 5. Hence, the correlation coefficient with the new combined group consisting of objects 2 and 5 is 0.900.

1. Group A represents three objects and group B one object. The Euclidean distance of group A to group C is 4.53 and group B to group C is 7.81. What is the unweighted distance of the combined groups A and B to group C?
   (a) 6.17
   (b) 6.99
   (c) 5.35
   (d) Cannot say unless we know the number of objects in group C.

4.8.3 Next Steps

The next steps consist of continuing to group the data just as discussed above, until all objects have joined one large group. As there are six original objects, there will be five steps before achieving this. At each step, the most similar pair of objects or clusters is identified, and then they are combined into one new cluster, until all objects have been joined. The calculation is illustrated in Table 4.20, using nearest neighbour linkage, with the most similar objects at each step indicated in bold, and the new similarity measures shaded. In this particular example, all objects ultimately belong to the same cluster, although arguably object 1 (and possibly 3) does not have a very high similarity to the main group. In some cases, several clusters can be formed, although ultimately one large group is usually formed.

It is normal to then determine at what similarity measure each object joined a larger group, and so which objects resemble each other most.

1. How many linkage steps are there if clustering 20 objects?
   (a) 20
   (b) 19
   (c) Cannot say

4.8.4 Dendrograms

Often, the result of hierarchical clustering is presented in the form of a dendrogram (sometimes called a *tree diagram*). The objects are organised in a row, according to their similarities: the vertical axis represents the similarity measure at which each successive object joins a group. Using nearest neighbour linkage and correlation coefficients for similarities, the dendrogram in Table 4.20 is presented in Figure 4.28. It can be seen that object 1 is very different from the others. In this case, all the other objects appear to form a single group, but other clustering methods may give slightly different results. A good approach is to perform several different methods of cluster analysis and compare the results. If similar clusters are obtained, no matter which method is employed, we can rely on the results. Note that clusters can sometimes be reflected; for example, a cluster that is on the right side using one type of software may be on the left side using another but represent the same result, as there is no rule as to whether new branches form to the right or left.

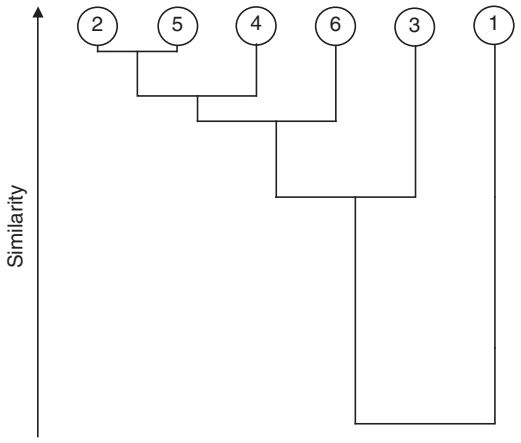


Figure 4.28 Dendrogram for cluster analysis example.

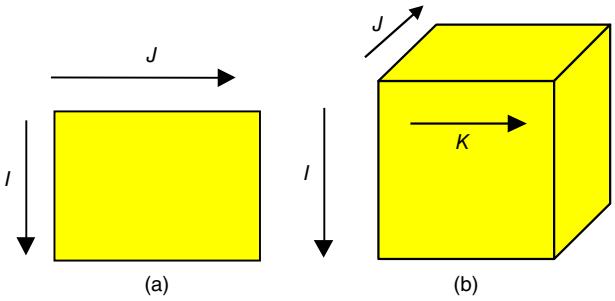


Figure 4.29 Two-way and three-way data.

1. A dendrogram will always have the same order of objects if the original data, linkage method and distance metric are the same.
   (a) True
   (b) False

4.9 Multi-way Pattern Recognition

Most traditional chemometrics is concerned with two-way data, often represented by matrices. Yet, over the past few years, a large interest in three-way chemical data has grown. Instead of organising the information as a two-dimensional array (Figure 4.29(a)), it falls into a three-dimensional 'tensor' or box (Figure 4.29(b)). Such data sets are surprisingly common. In Chapter 6, we will discuss multi-way partial least squares (PLS) (Section 6.5.3), the discussion in this section being restricted to EDA.

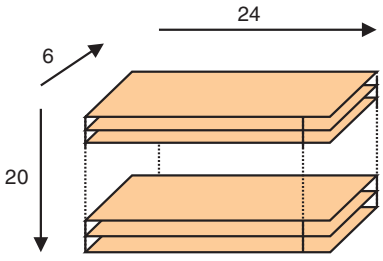
Consider, for example, an environmental chemical experiment in which the concentrations of six elements are measured at 20 sampling sites on 24 days in a year. There will be $20 \times 24 \times 6$ or 2880 measurements; however, these can be organised as a 'box' with 20 planes each corresponding to a sampling site and of dimensions 24×6 (Figure 4.30). Such data sets have been available for many years to psychologists and in sensory research. A typical example might involve a taste panel assessing 20 food products. Each food could involve the use of 10 judges who score eight attributes, resulting in a $20 \times 10 \times 8$ box. In psychology, we might be following the reactions of 15 individuals to five different tests on 10 different days, possibly each day under slightly different conditions, hence have a $15 \times 5 \times 10$ box. These problems involve finding the main factors that influence the taste of a food or the source of pollutant or the reactions of an individual and are a form of pattern recognition.

Three-dimensional analogies to principal components are required. There are no direct analogies to scores and loadings as in PCA; hence, the components in each of the three dimensions are often called *weights*. A number of methods are available to tackle this problem.

4.9.1 Tucker3 Models

These models involve calculating weight matrices corresponding to each of the three dimensions (e.g. sampling site, date and metal), together with a 'core' box or array, which provides a measure of magnitude. The three weight matrices

Figure 4.30 Possible method of arranging environmental sampling data.



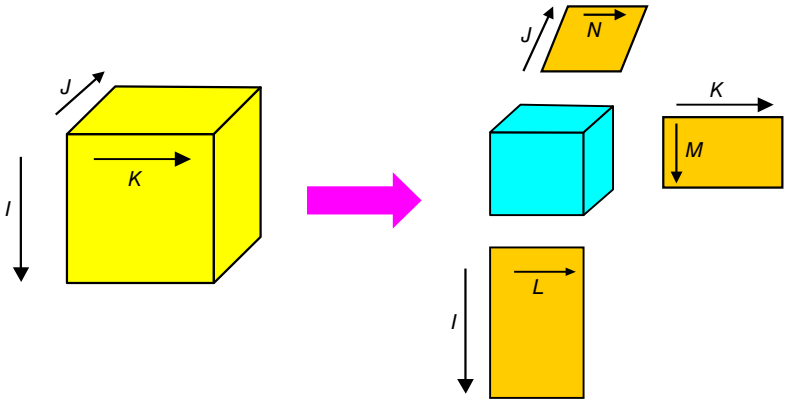


Figure 4.31 Tucker3 decomposition.

do not necessarily have the same dimensions; hence, for example, the number of significant components for the sampling sites may be different to those for the dates, unlike normal PCA where one of the dimensions of both the scores and loadings matrices must be identical. This model (or decomposition) is represented in Figure 4.31. The easiest mathematical approach is by expressing the model as a summation

$$x_{ijk} \approx \sum_{l=1}^{L} \sum_{m=1}^{M} \sum_{n=1}^{N} a_{il} b_{jm} c_{kn} z_{lmn}$$

where z represents what is often called a *core array* and a, b and c are functions relating to each of the three types of variables. Some authors use the concept of 'tensor multiplication' being a 3D analogy to 'matrix multiplication' in two dimensions; however, the details are confusing, and it is conceptually probably best to stick to summations, which computer programs do.

1. For a Tucker3 model, all dimensions of the core array must be equal.
   (a) True
   (b) False

4.9.2 Parallel Factor Analysis (PARAFAC)

Parallel factor analysis (PARAFAC) differs from the Tucker3 models in which each of the three dimensions contains the same number of components. Hence, the model can be represented as the sum of contributions due to g components, just as in normal PCA, as illustrated in Figure 4.32 and represented algebraically by

$$x_{ijk} \approx \sum_{g=1}^{G} a_{ig} b_{jg} c_{kg}$$

Each component can be characterised by one vector that is analogous to a scores vector and two vectors that are analogous to loadings, but some keep to the notation of 'weights' in three dimensions. Components can, in favourable

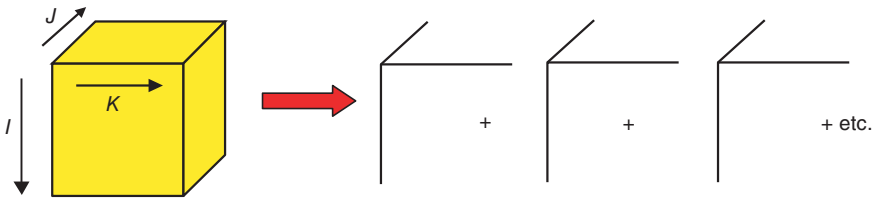


Figure 4.32 Parallel factor analysis (PARAFAC).

circumstances, be assigned a physical meaning. A simple example might involve following a reaction by recording a diode array HPLC chromatogram at different reaction times. A box whose dimensions are *reaction time* × *elution time* × *wavelength* can be used to represent the data. If there are four factors in the data, this would imply four significant compounds in a cluster in the HPLC (or four significant reactants), and the weights should correspond to the reaction profile, the chromatogram and the spectrum of each compound. Three-way methods in chemometrics had a special origin in fluorescence spectroscopy where two of the dimensions are emission and excitation wavelengths, which are quite stable spectroscopically. In chromatography, there may be difficulties aligning the chromatographic dimension, making PARAFAC somewhat less useful.

PARAFAC, however, is quite difficult to use and, although the results are easy to interpret physically, it is conceptually more complex than PCA. It can, however, lead to results that are directly interpretable, whereas the factors in PCA have a purely abstract meaning.

1. The dimensions of the weights in a PARAFAC model must be the same.
   (a) True
   (b) False

4.9.3 Unfolding

Another approach is to simply 'unfold' the 'box' to give a long matrix. In the environmental chemistry example, instead of each sample being represented by a 24 × 6 matrix, it could be represented by a vector of length 144, each element consisting of the measurement of one element on one date, for example, the measurement of Cd concentration on July 15. Then, a matrix of dimensions 20 (sampling sites) × 144 (variables) is produced (Figure 4.33) and subjected to normal PCA. Note that a box can be subdivided into planes in three different ways (compare Figure 4.30 with Figure 4.33), according to which dimension is regarded as the 'major' dimension. While unfolding, it is also quite important to consider details of scaling and centring, which become far more complex in three dimensions as opposed to two. After unfolding, normal PCA can be performed. Components can be averaged over related variables; for example, we could take an average loading for Cd over all dates to give an overall picture of its influence on the observed data.

This comparatively simple approach is sometimes sufficient, but the PCA calculation neglects to take into account some relationships between the variables. For example, the relationship between concentration of Cd on July 15 and that on August 1, in an environmental analysis, is considered to be no stronger than the relationship between Cd concentration on July 15 and Hg on November 1 during the calculation of the components. However, after the calculations are performed, it is still possible to regroup the loadings and sometimes an easily understood method such as unfolded PCA can be of value.

1. Forty mice are assayed daily over 2 weeks for the concentrations of 20 compounds. The data can be unfolded into a matrix of dimensions
   (a) 40 × 280
   (b) 40 × 20
   (c) 40 × 14 × 20

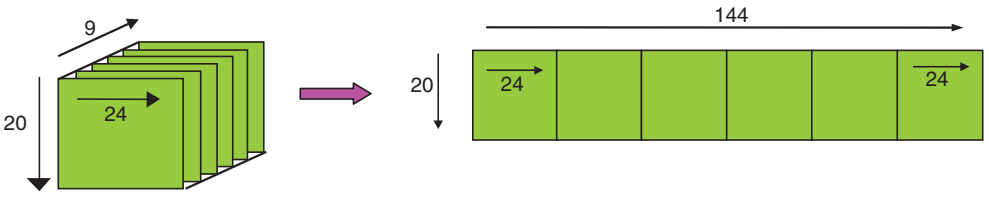


Figure 4.33 Unfolding.